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I want to comment on the last question of the quiz, since it is of fundamental importance. The result is that the upper left block entry of Σ^{-1} is exactly equal to $\text{Var}(X_1 | X_2)$.

Thus, by label invariance if X_a is any subvector of X and X_b is the remaining, the entries of Σ^{-1} are $\text{Var}(X_a | X_b)$. (This can be seen simply by moving X_a to the first subvector of X , which can't change the conditional distribution.)

Also recall in multivariate normals that lack of covariance is equivalent to independence.

The result is that Σ^{-1} gives all of the information about the conditional independence structure of subvectors of X . So, for example, if $X = [x_1 \ x_2 \ x_3 \ \dots]'$ (all one dimensional). Then to find out if X_i is independent of X_j given all of the other elements of X simply look at whether the i, j element of Σ^{-1} is zero or not.

This simple result forms much of the basis of so-called Gaussian graphical models.

[Mark as completed](#)

